

Week 6

NICHD RNAseq Course, Spring 2025

March 4th, 2025

We are here!

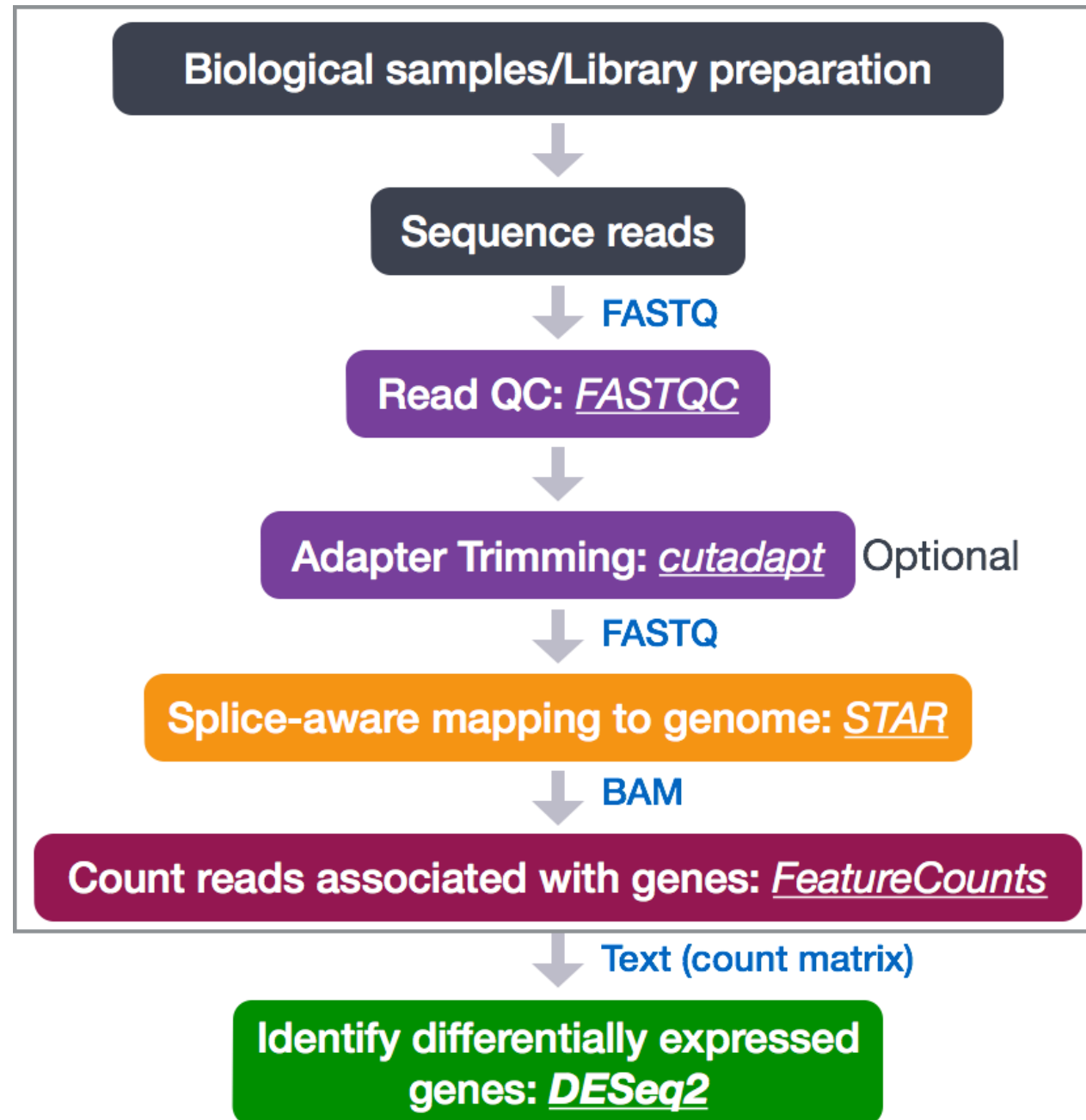
Geneid	Irrel_kd_1	Irrel_kd_2	Irrel_kd_3	Mov10_oe_1	Mov10_oe_2	Mov10_oe_3
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	82	86	65	119	106	52
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	2	1	0	3	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000001	0	0	0	0	0	0
ENSG000002	2	0	0	2	1	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	1	1	0	0	1	0

samples: want to see if differences across condition are significant (w.r.t. biological and technical variation)

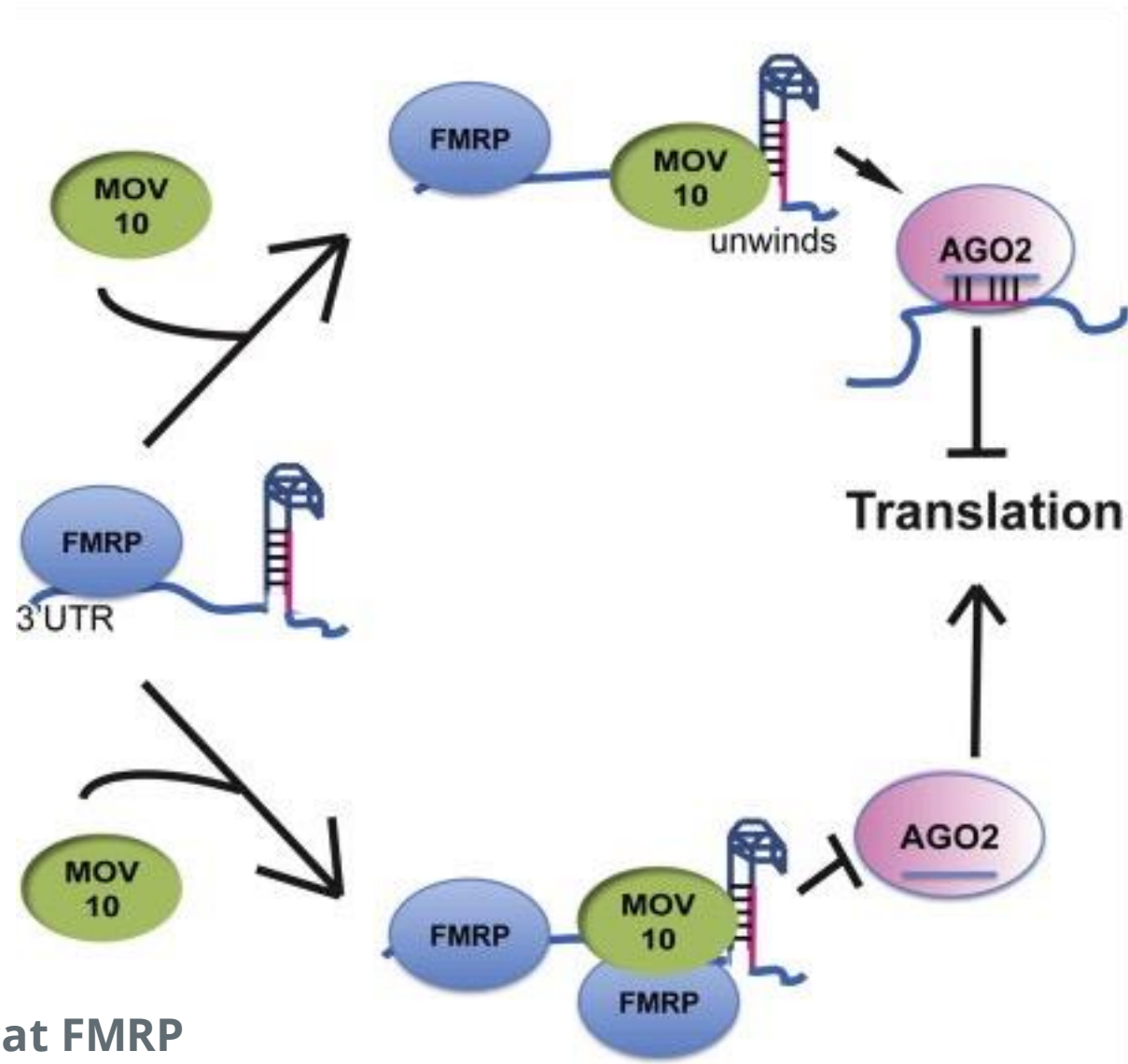
features (e.g. genes)

ENSG000000000003
ENSG000000000005
ENSG000000000419
ENSG000000000457
ENSG000000000460

SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
679	448	873	408	1138
0	0	0	0	0
467	515	621	365	587
260	211	263	164	245
60	55	40	35	78



What is our example dataset testing?



The hypothesis [the paper](https://nichd-bspc.github.io/intro-rnaseq-hpc/lessons/wk6_lesson01_dge_setup_overview.html) is testing is that FMRP and MOV10 associate and regulate the translation of a subset of RNAs.

https://nichd-bspc.github.io/intro-rnaseq-hpc/lessons/wk6_lesson01_dge_setup_overview.html

What is our example experimental design?

Irrelevant kd (knock down)

Replicate1

Replicate2

Replicate3

Mov10 kd (knock down)

Replicate2

Replicate3

Mov10 oe (over expression)

Replicate1

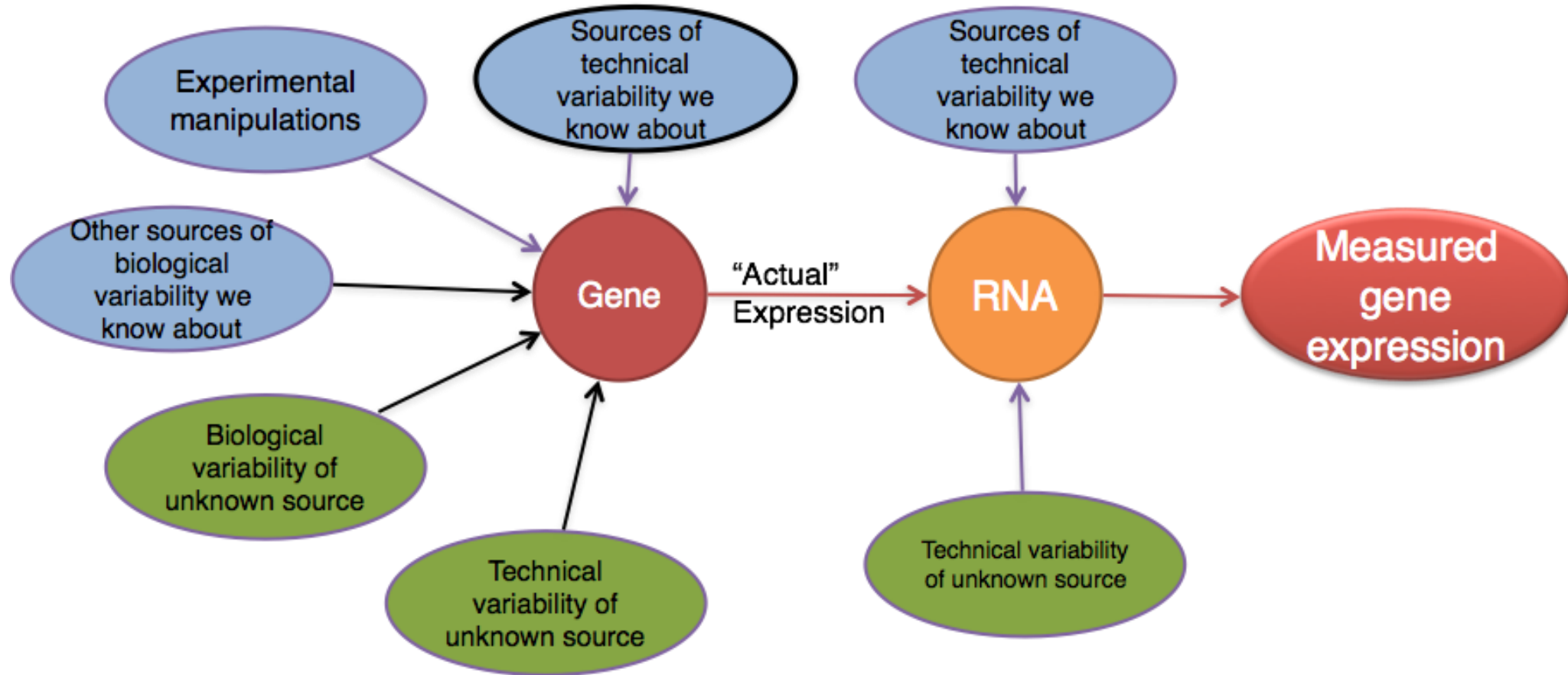
Replicate2

Replicate3

The RNA-Seq was performed on HEK293F cells that were either transfected with a MOV10 transgene, or siRNA to knock down Mov10 expression, or non-specific (irrelevant) siRNA.

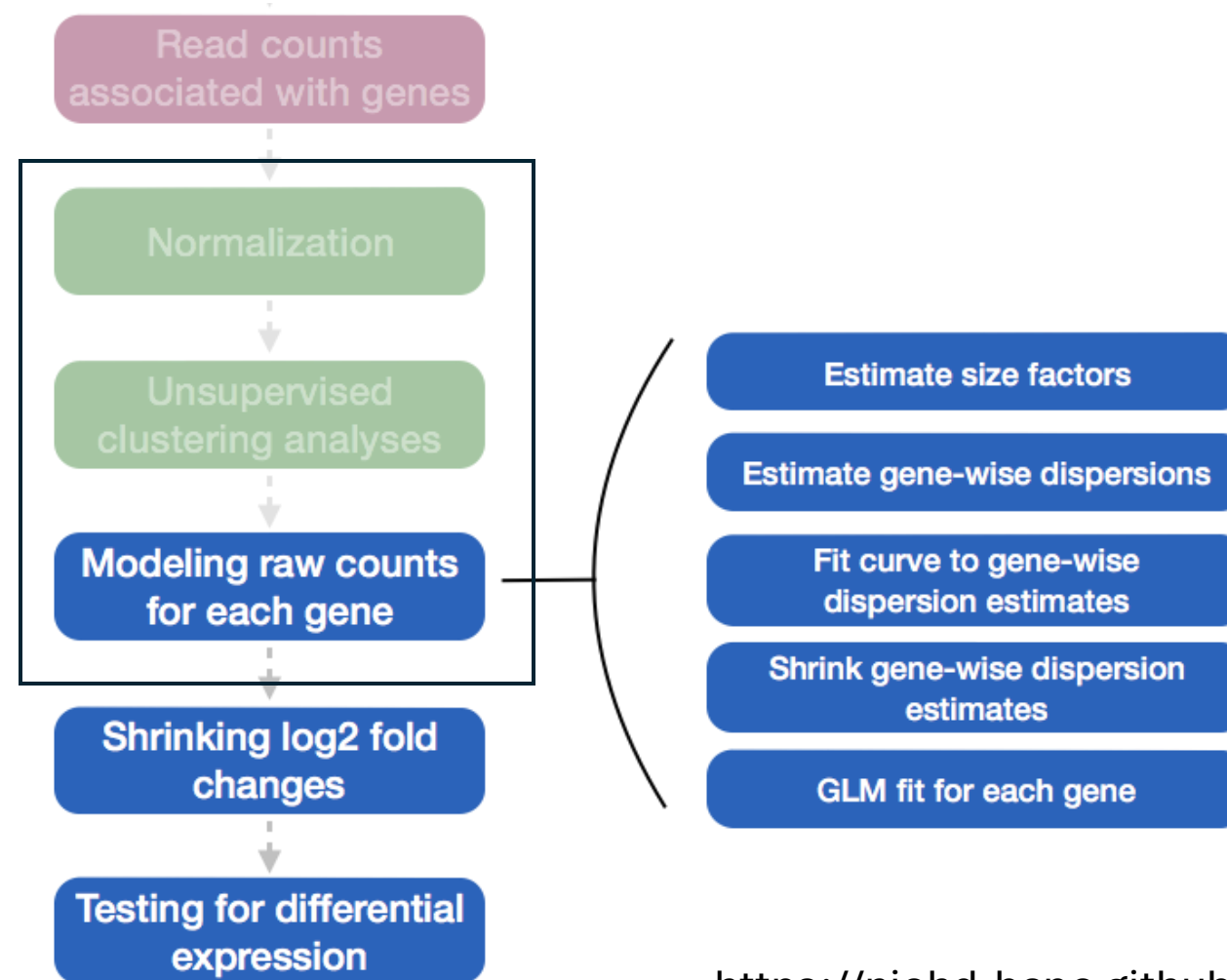
https://nichd-bspc.github.io/intro-rnaseq-hpc/lessons/wk6_lesson01_dge_setup_overview.html

It is never just as simple as our design



Courtesy of Paul Pavlidis, UBC

DESeq2 pipeline takes a lot of this into account



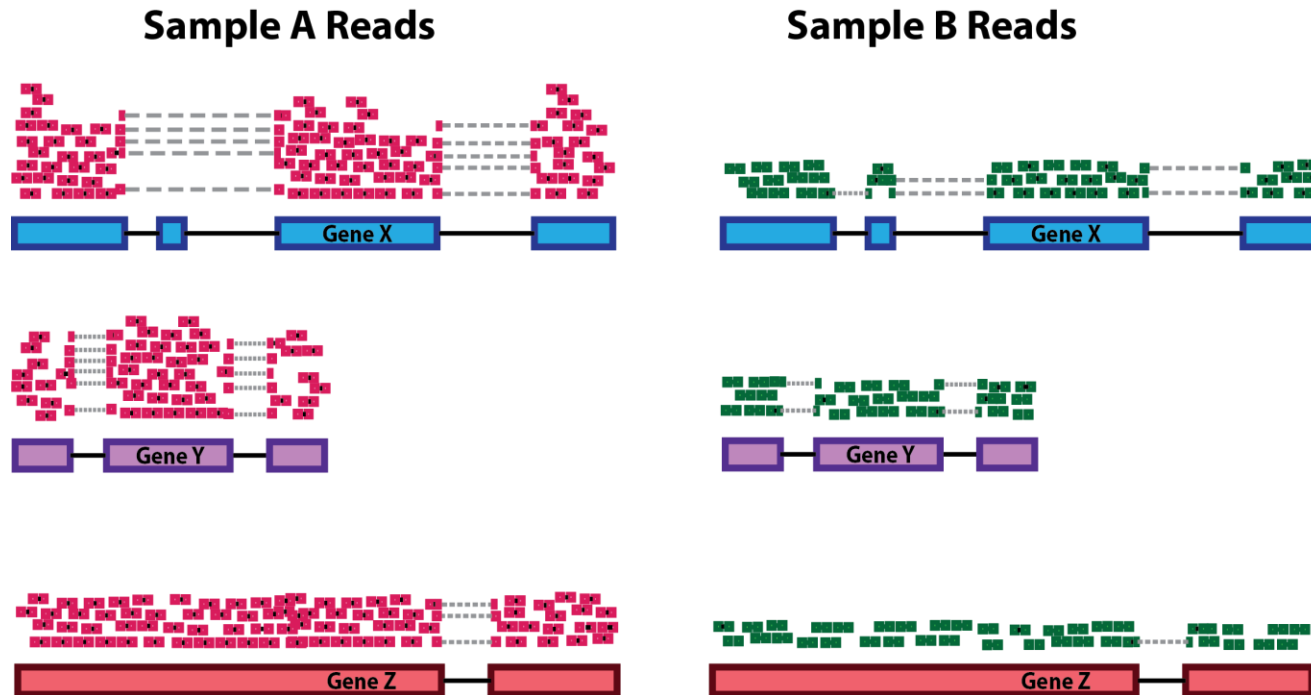
https://nichd-bspc.github.io/intro-rnaseq-hpc/lessons/wk6_lesson01_dge_setup_overview.html

Today:

- Loading data into RStudio and DEseq2
- Loading required packages for data viz and analysis
- Understanding the underlying distribution of read counts
- Normalization
- If time: Sample-level QC with Principal components analysis

Normalization

https://hbctraining.github.io/DGE_workshop_salmon/lessons/02_DGE_count_normalization.html



All genes in sample A appear to be 2x the expression of sample B

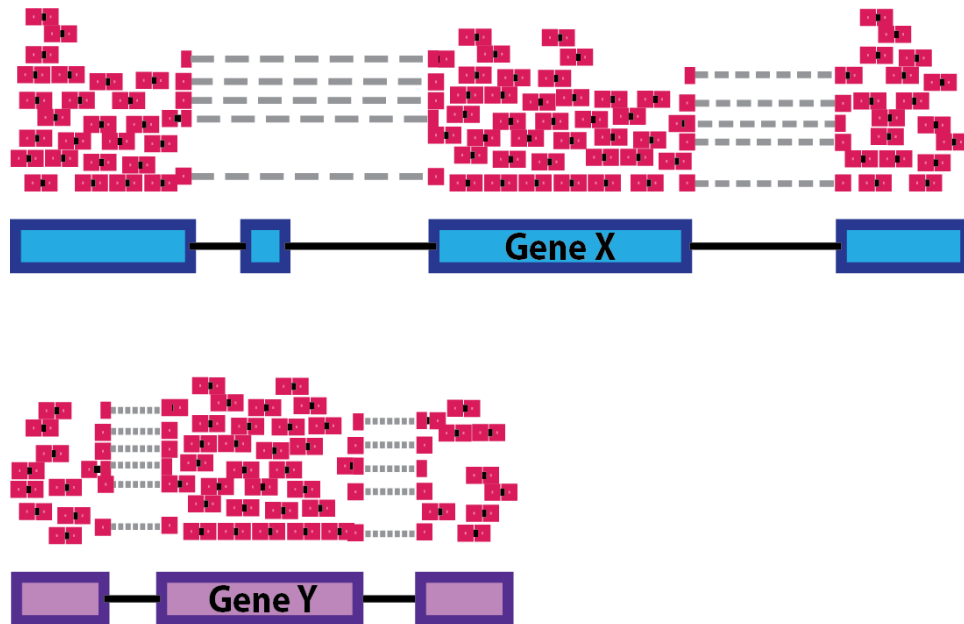
This only due to sample A being sequenced 2x deeper.

So we will need to **normalize to depth**

Normalization

https://hbctraining.github.io/DGE_workshop_salmon/lessons/02_DGE_count_normalization.html

Sample A Reads



Within one sample, a single transcript of a long gene gives more reads than one transcript of a short gene. That's because long transcripts get fragmented into many pieces, and we are counting pieces.

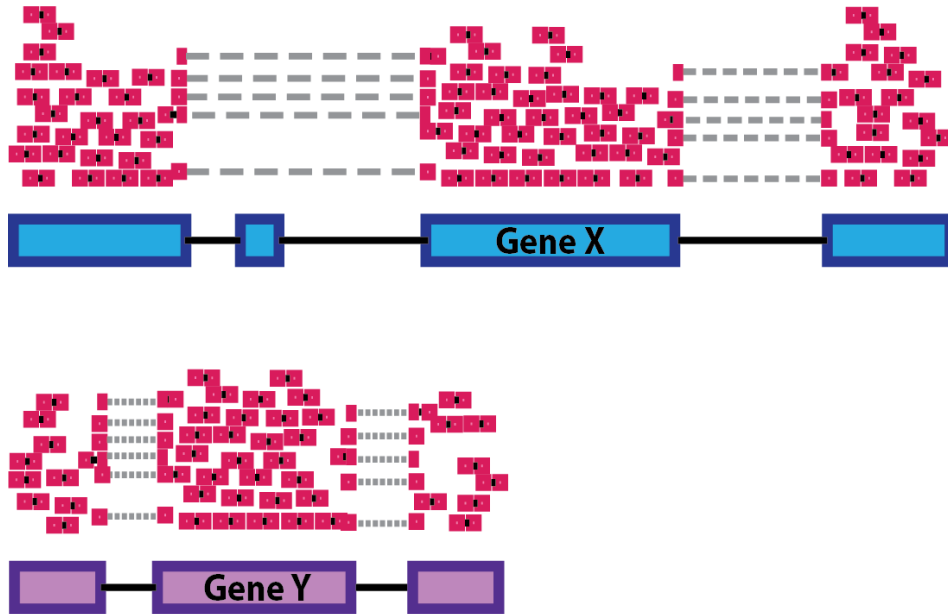
When comparing ***within*** a sample, we need to **normalize by gene length**.

When comparing *across* samples, we assume gene lengths do not change due to condition. So we do not need length normalization.

Normalization

https://hbctraining.github.io/DGE_workshop_salmon/lessons/02_DGE_count_normalization.html

Sample A Reads

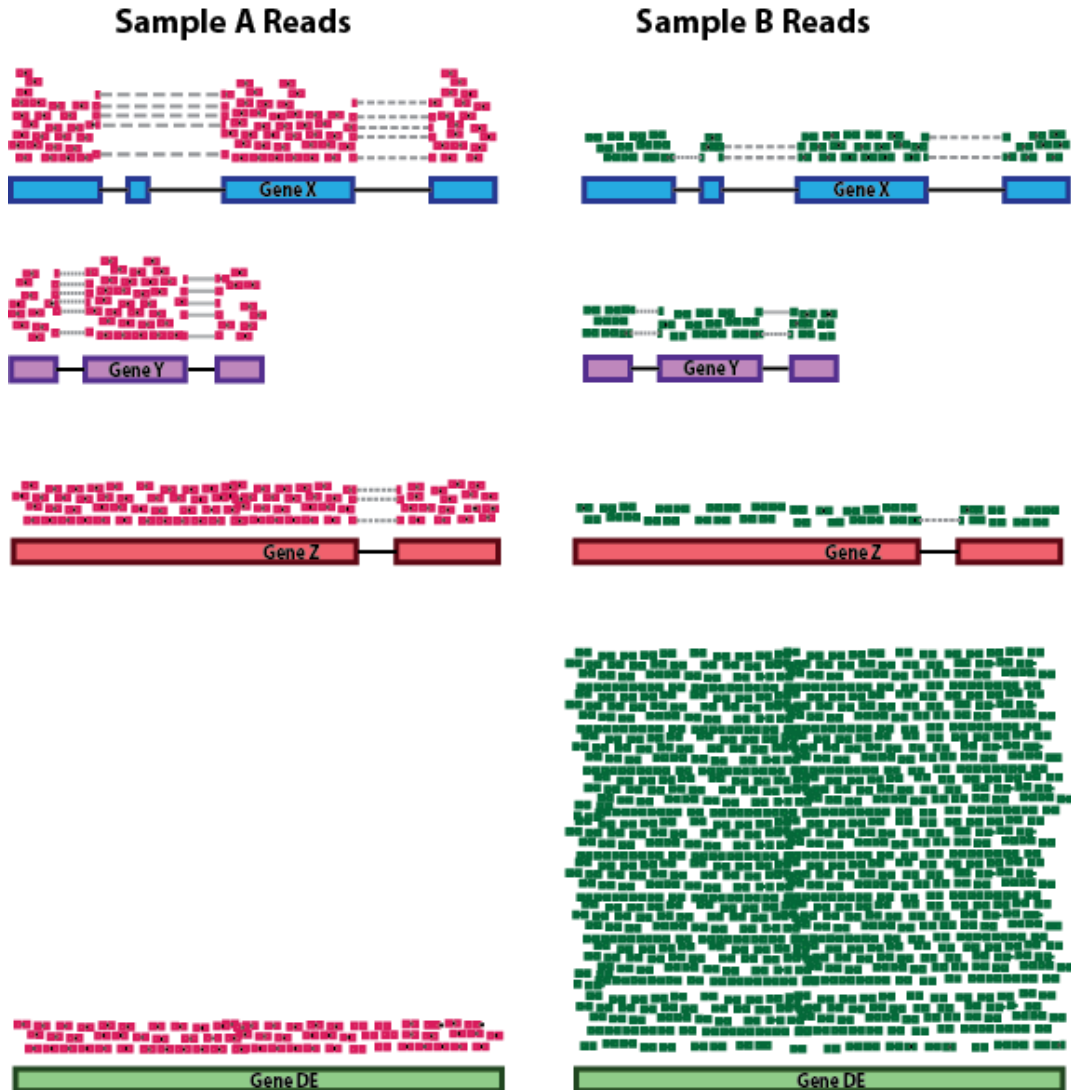


This bears repeating:

In typical DE analysis, **counts are not normalized to gene length** because we assume length does not change across our samples.

Normalization

https://hbctraining.github.io/DGE_workshop_salmon/lessons/02_DGE_count_normalization.html



- A sequencer's flow cell has a finite number of reads.
- One highly-expressed gene “soaks up” the reads in sample B, causing other genes to look lower than they are.
- This difference in *composition* needs to be taken into account.

Side note: why is everything on a log scale?

- RNA expression levels span 6 orders of magnitude (one transcript per cell to millions of transcripts in a cell) and distribution can be very skewed.
- When expression increases, so does variance. This is a problem for statistical methods (heteroskedasticity)
- Fold changes (the preferred method of quantifying differential expression) are not symmetric around 0, which is also a problem for statistical operations
- Log-transformed data solves all of this

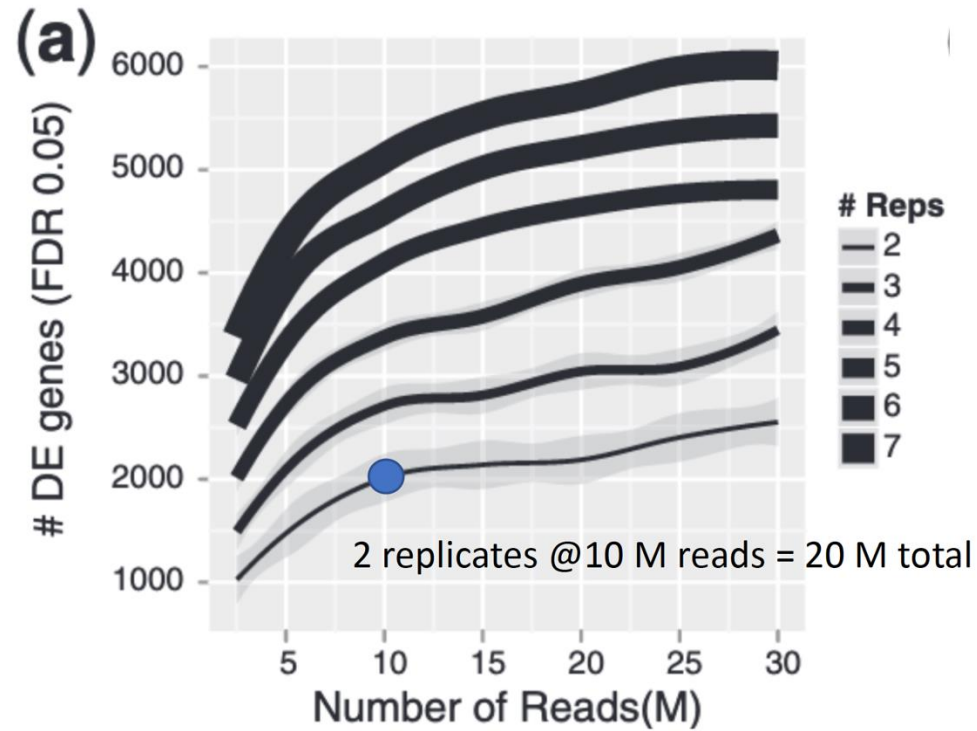
Replicates vs depth

For \$2500, you get ~400M reads. How do you best allocate those reads?

- Sequencing deeper:
 - catch low-expression genes
 - improve signal overall
- More replicates:
 - improve statistical power

You can't do both. Which strategy gives the most overall power (ability to detect truly differentially expressed genes)?

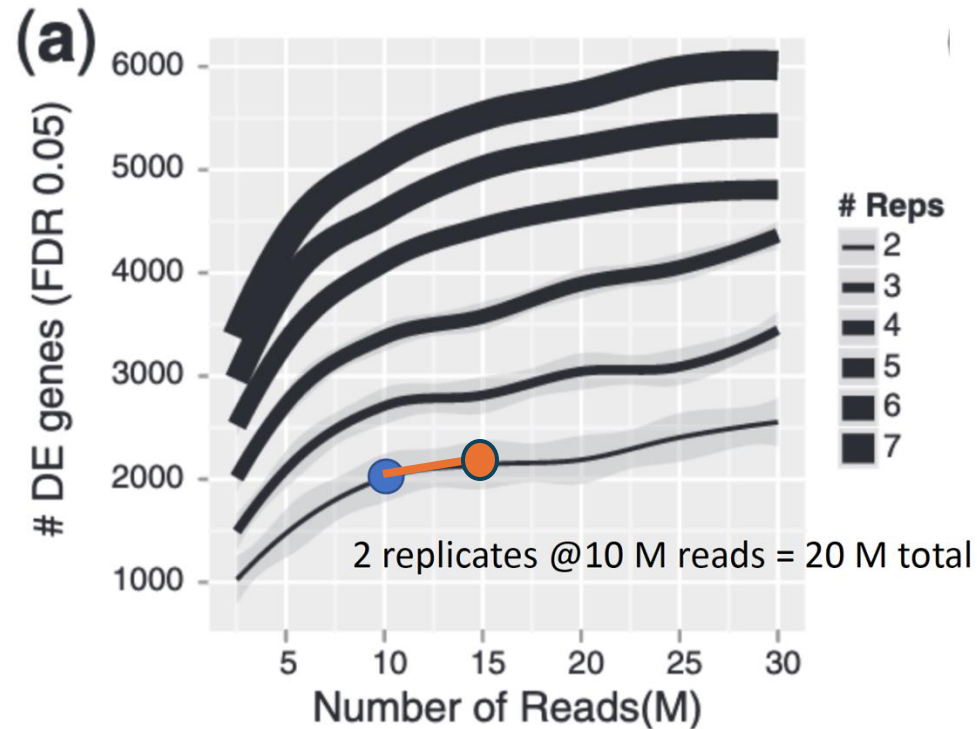
Always use more samples, not more depth



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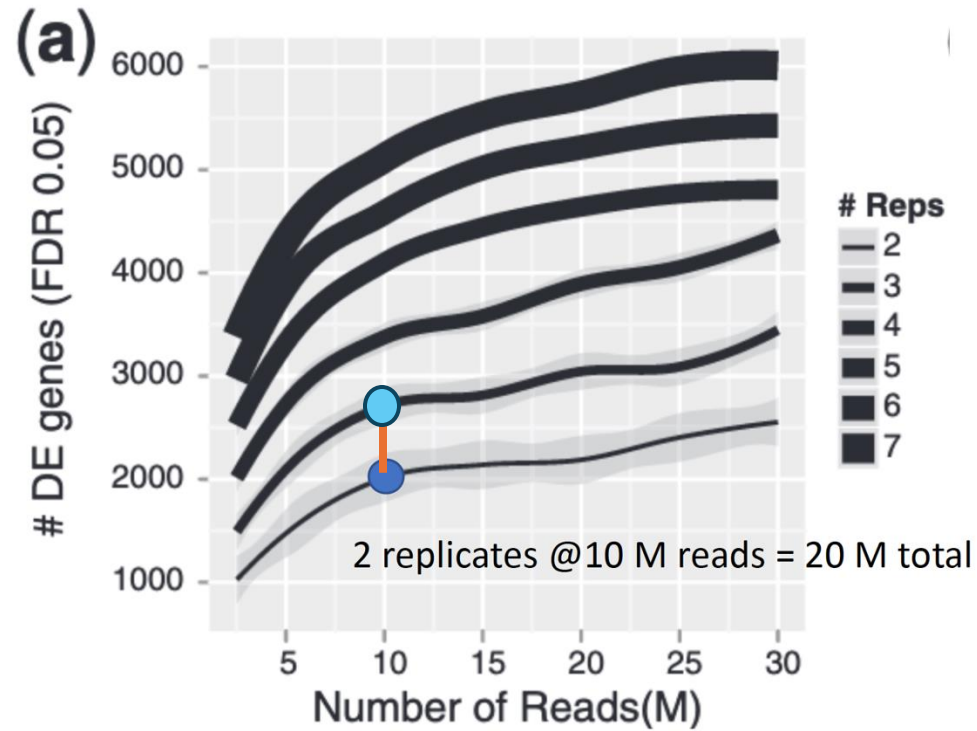
Always use more samples, not more depth

Add 10M reads by
adding 5M for each of
the 2 replicates



Always use more samples, not more depth

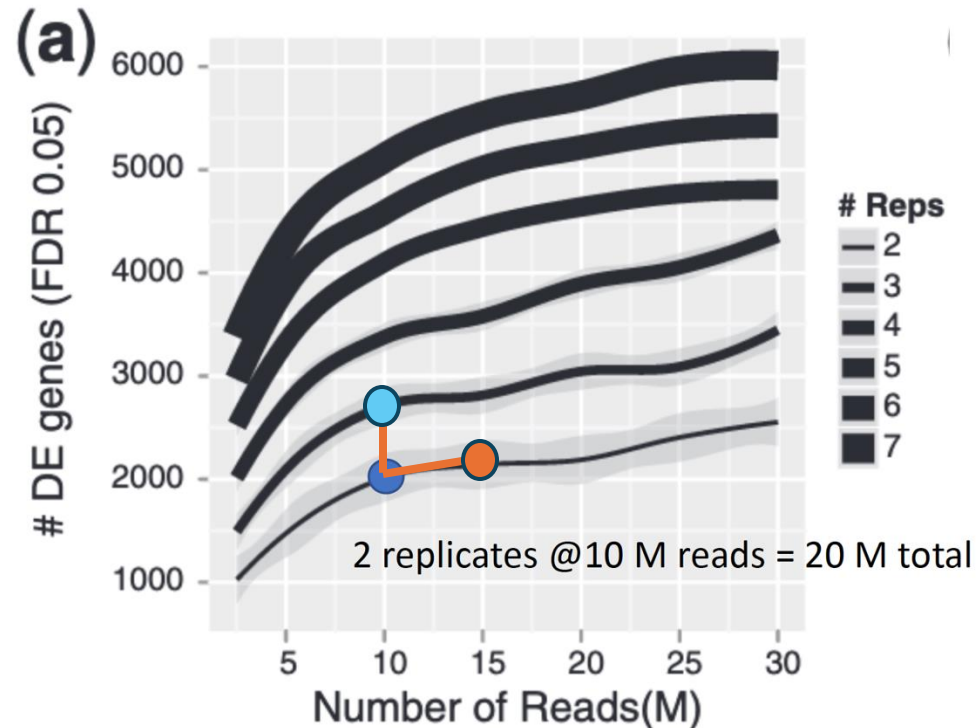
Add 10M reads by
adding another 10M
read sample



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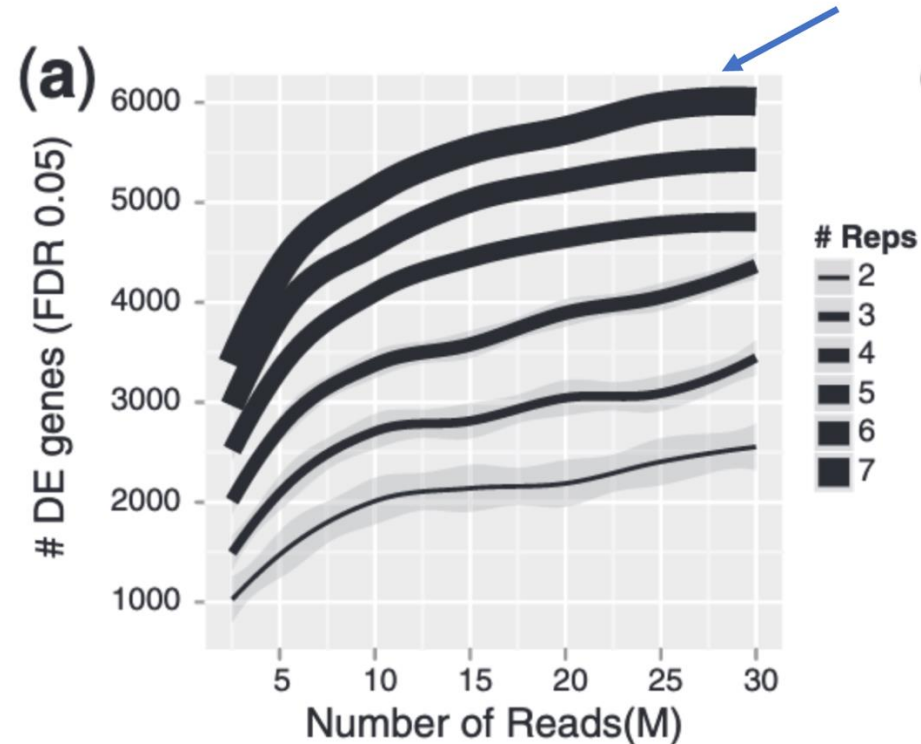
Always use more samples, not more depth

- Another sample increases power by 700 DE genes.
- More depth only increases by 50 DE genes.



Always use more samples, not more depth

DE detection does not saturate!

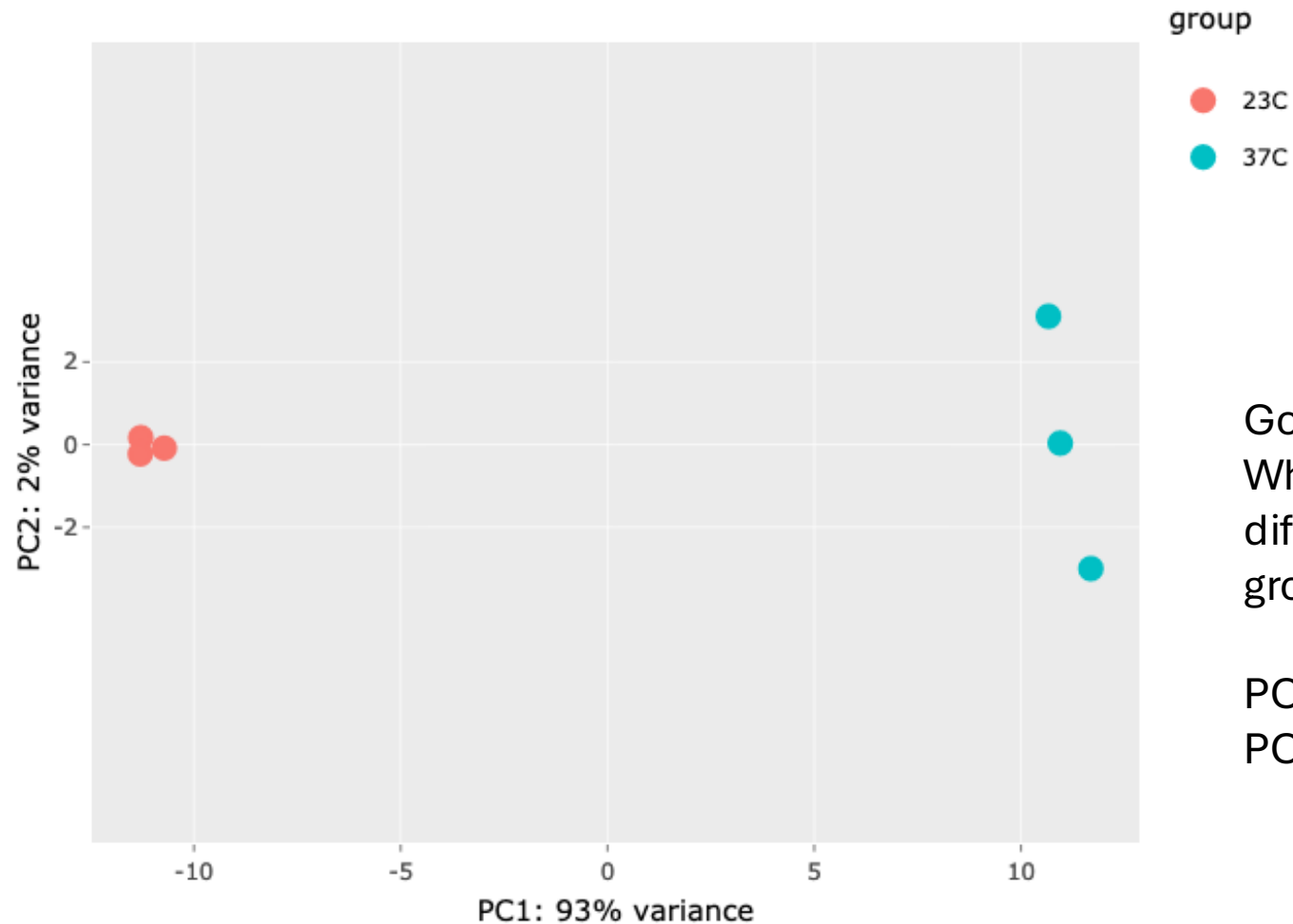


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Sample similarity (PCA plots)

- Principal Components Analysis (PCA) is a dimension reduction technique used in machine learning and any time you need to summarize multi-dimensional data
- No time for math today, will focus on visual aspect
- PCA worked example:
https://hbctraining.github.io/DGE_workshop_salmon/lessons/03_DGE_QC_analysis.html
- StatQuest video:
https://www.youtube.com/watch?v=_UVHneBUBW0

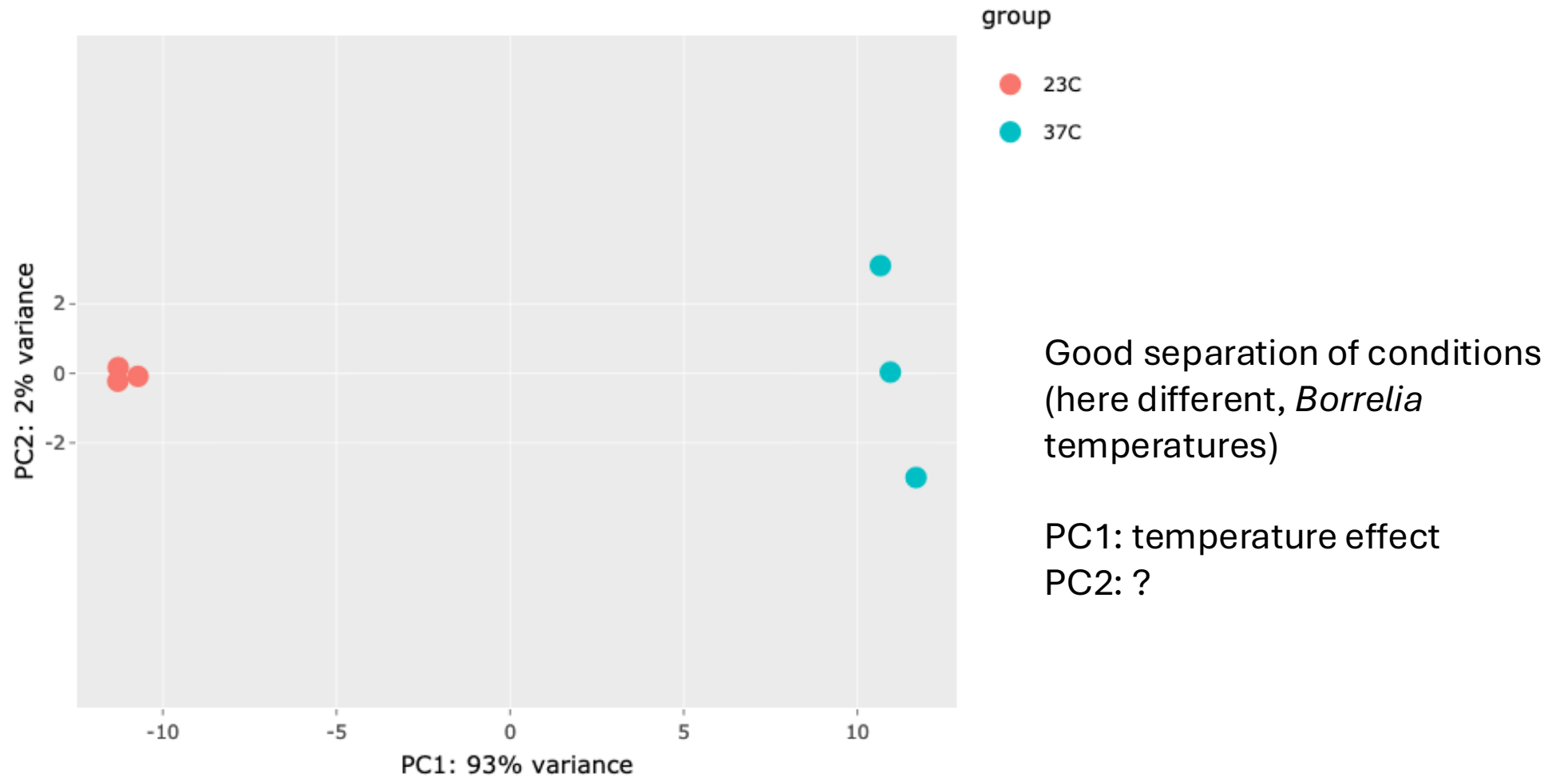
Two-condition PCA plot



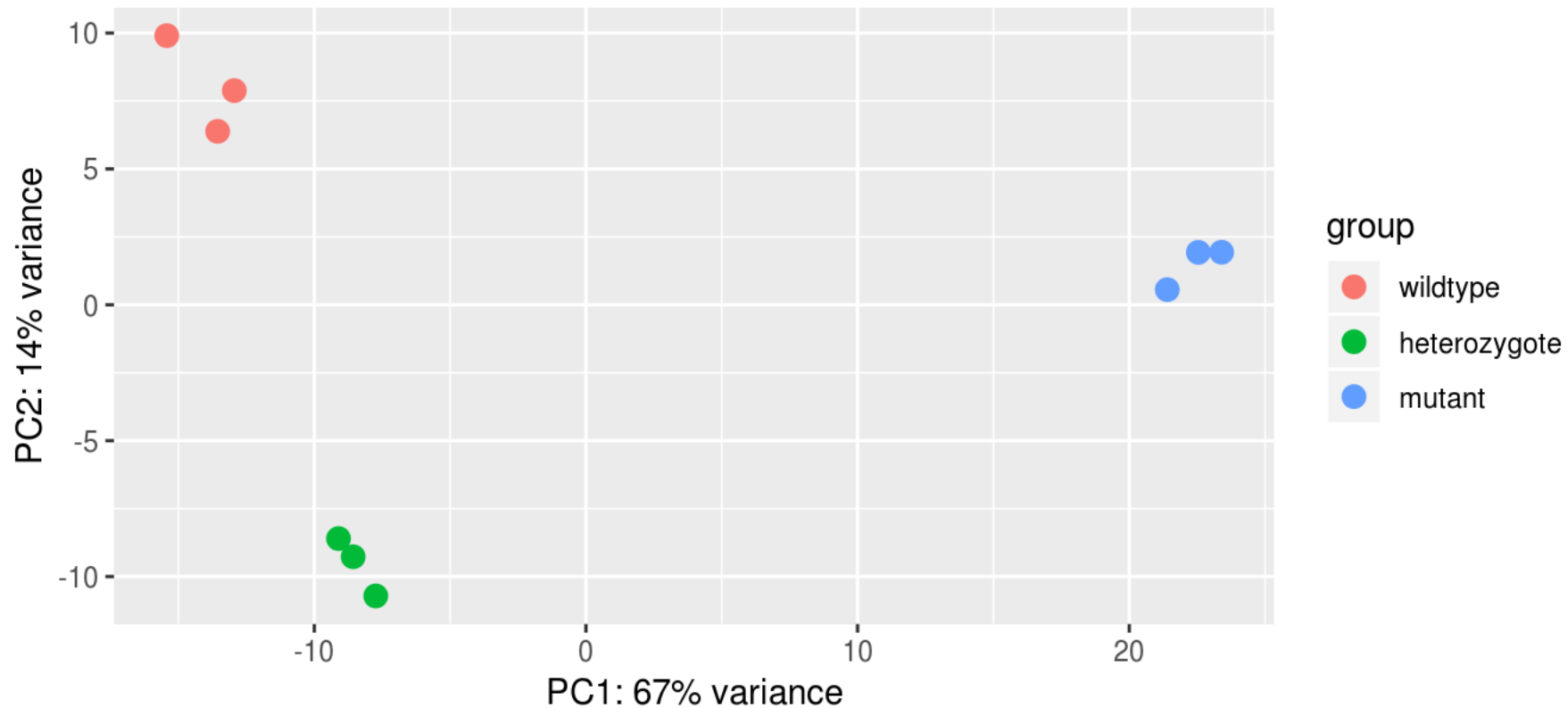
Good separation of conditions
Which PC corresponds to the
difference between our sample
groups?

PC1:
PC2:

Two-condition PCA plot



Three-condition PCA plot



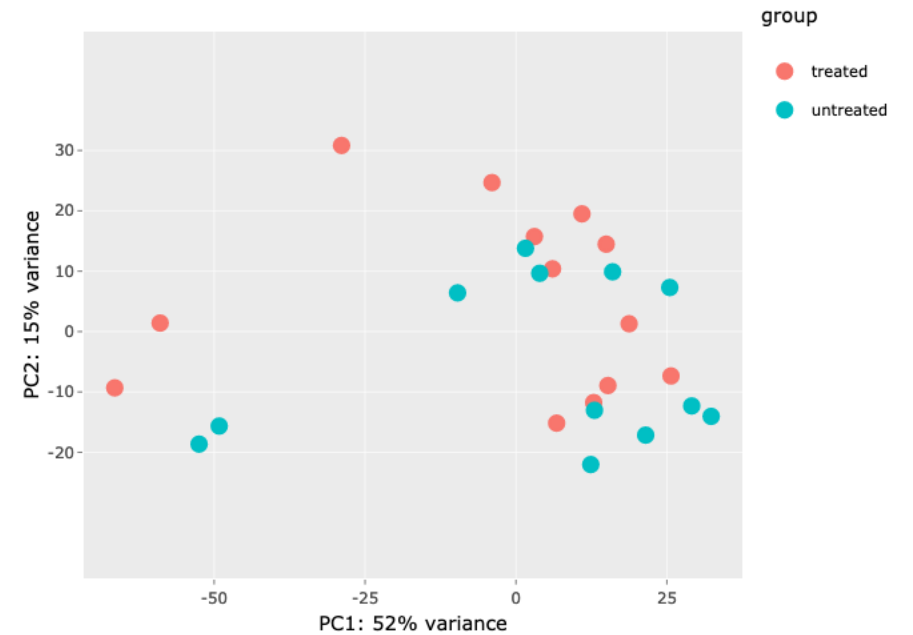
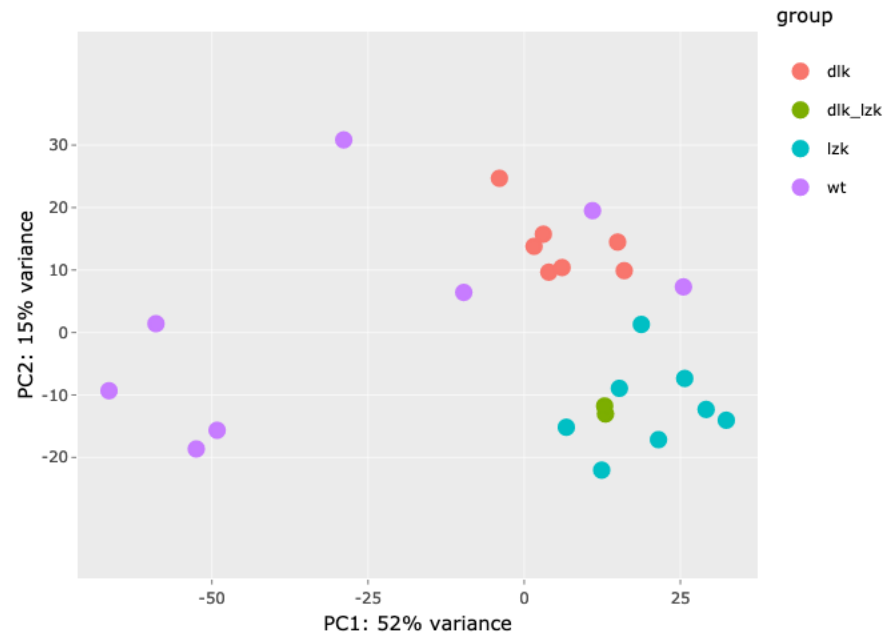
We might hypothesize that the het is more similar to WT than it is to the mutant (PC1)

But it's flipped when we look at PC2, likely due to different genes than those separating the mutant from WT

See StatQuest video for context and help thinking about this:

https://www.youtube.com/watch?v=_UVHneBUBW0

Sometimes it gets crazy



Same plots, colored by different metadata (genotype or treatment)

Complicated experiment, hard to interpret PC1 and PC2

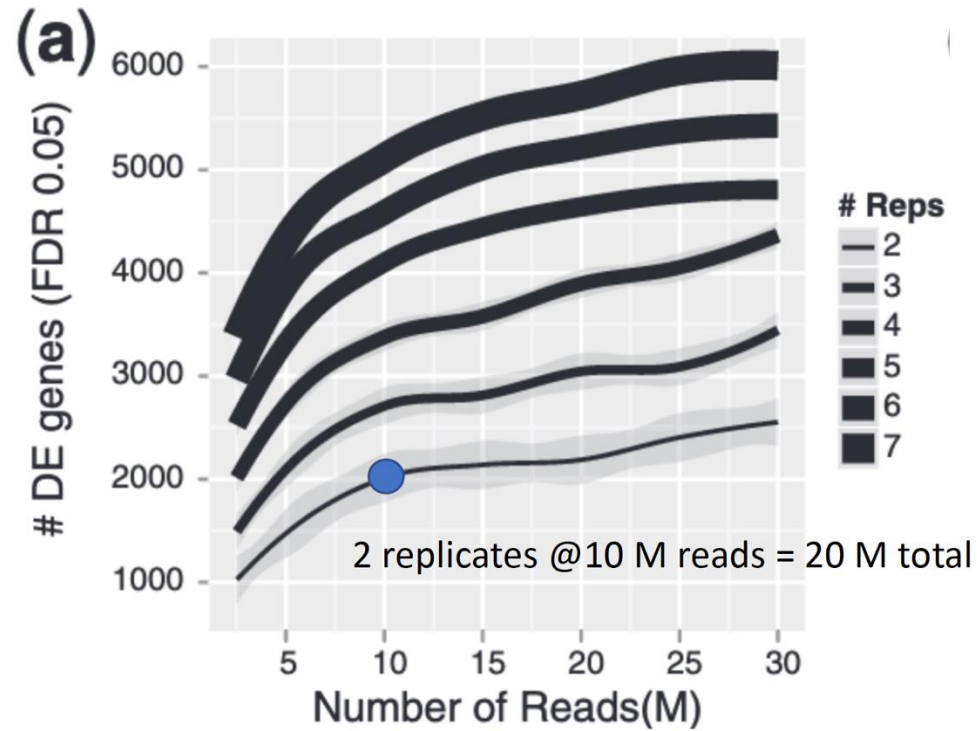
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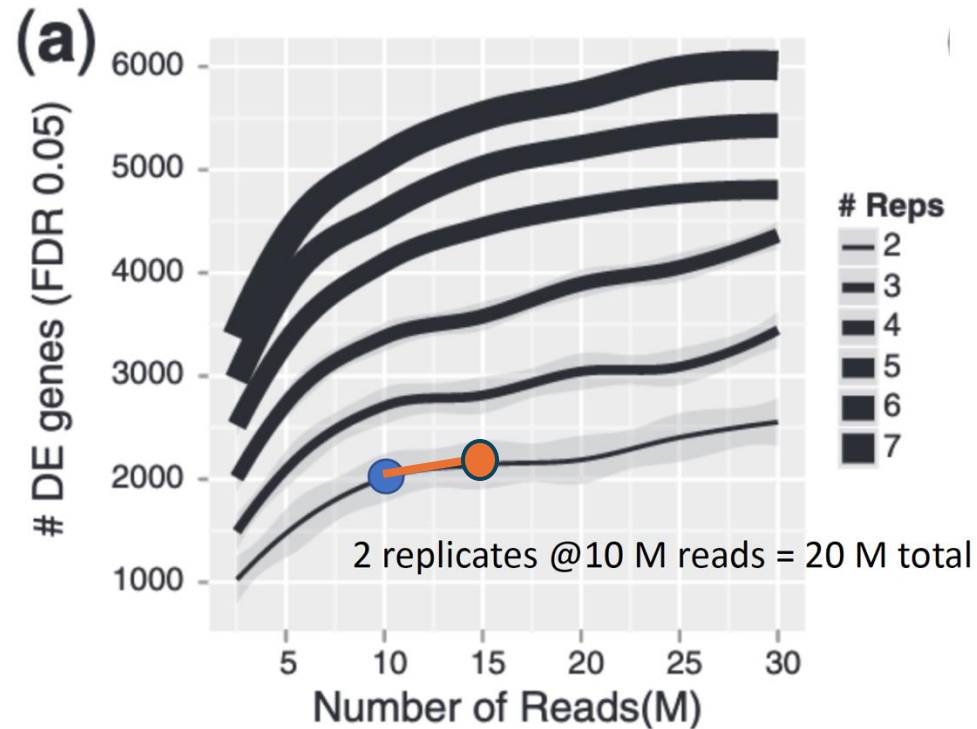
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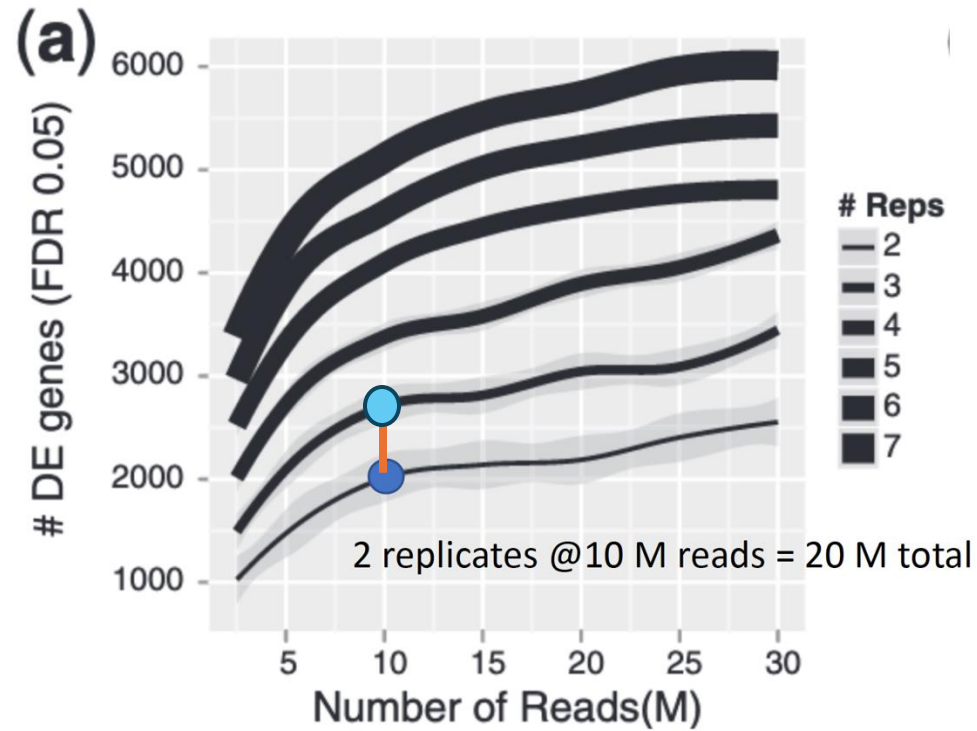
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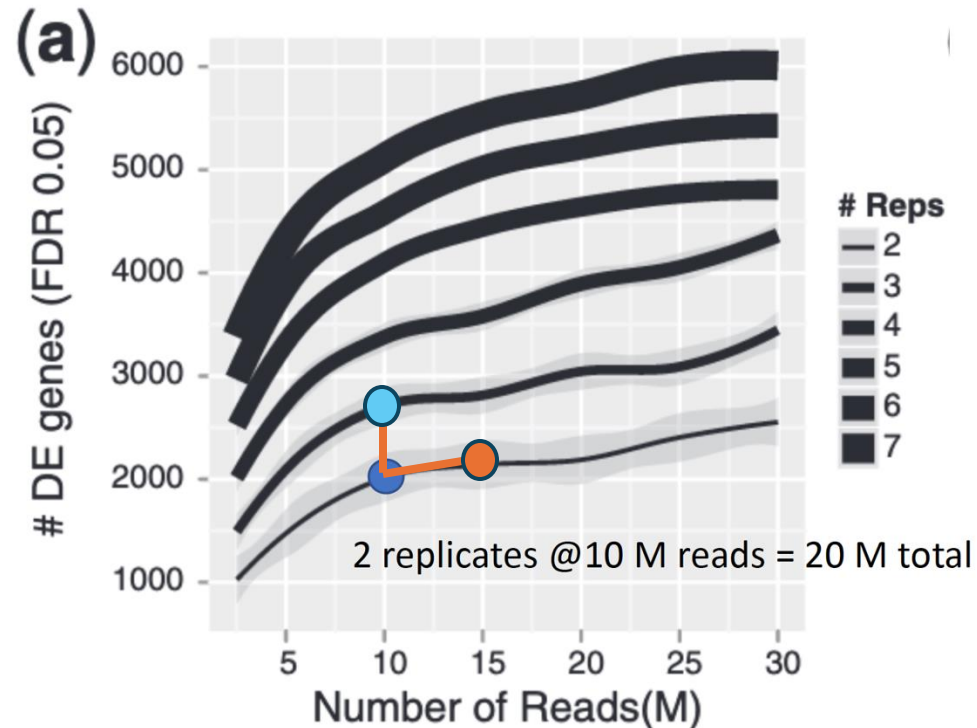
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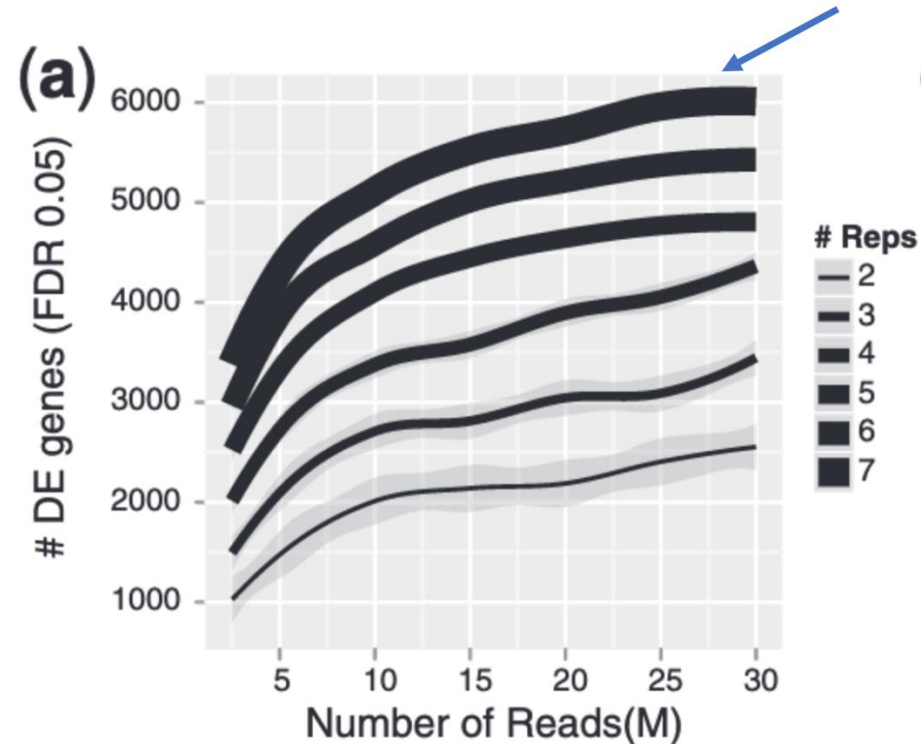
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We are skipping how DESeq2 works

